

Intro to Frequency Response

EGR 345

GVSU

Outline

- some thoughts on the projects
 - additional thoughts on on building vibration absorbers
- response of MSD system to sinusoidal input at various frequencies

Initial Project Thoughts

- things are really coming together
 - with your help

Building Vibration Absorbers

- step 1: building without absorber (no pendulum)
- step 2: passive pendulum
- step 3: magnetic pendulum damping
- step 4: active pendulum
 - pendulum attached to DC motor or servo
 - feedback control to create motor/servo input

What would Dr. Farris say about our projects?

- I think there is something he harps on
- I think there is a false dichotomy between ME and PDM

Analysis of building vibration absorbers

- what's wrong with trial and error?
 - I think I could make an argument from every major in the room

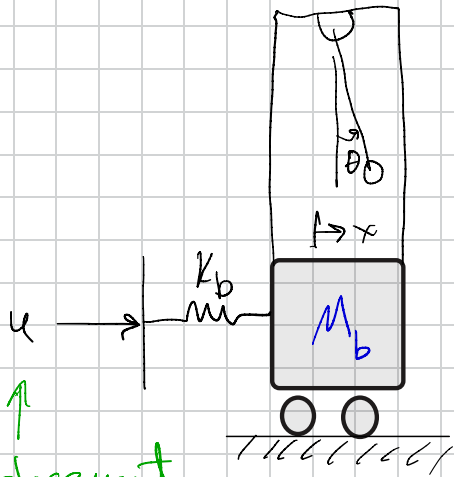
How do passive vibration absorbers work?

- and what role does damping play?

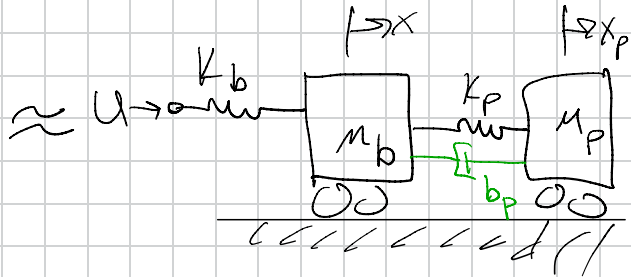
→ Not actuated

→ the pendulum or moving mass
is not driven by anything
- no motor, servo, or
whatever

Simple Earthquake model



↑
displacement
input

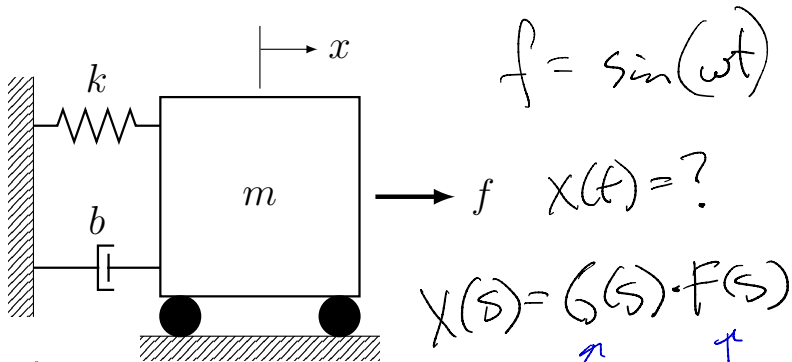


→ without pendulum
damping, it only
works @ 1 frequency

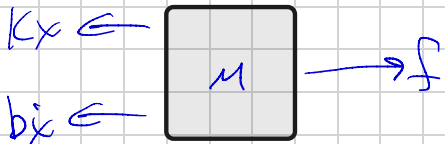
Intro to Frequency Response

How does an MSD system response to a sinusoidal input?

- low frequency, medium frequency, and high frequency



FBD $\rightarrow$ EOM $\xrightarrow{\mathcal{L}}$ $G(s) \cdot F(s) \rightarrow X(s) \xrightarrow{\text{PFE}} x(t) ?$



$$G(s) = \frac{1}{ms^2 + bs + k} \quad \frac{1}{m} \quad \frac{1}{m}$$

↑

$$m\ddot{x} = f - b\dot{x} - kx$$
$$m\ddot{x} + b\dot{x} + kx = f$$

$$X(s) = G(s) \cdot F(s)$$

$$(ms^2 + bs + k)X(s) = F(s) \rightarrow G(s) = \frac{X(s)}{F(s)}$$

$$G(s) = \frac{1}{ms^2 + bs + k} \cdot \frac{1}{\frac{1}{m}} = \frac{\frac{1}{m}}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

↑
1/m

$$X(s) = G(s) \cdot F(s)$$

$$f(t) = \sin(\omega t)$$

$$F(s) = \frac{\omega}{s^2 + \omega^2}$$

$$X(s) = \frac{\omega}{s^2 + \omega^2} \cdot \frac{1/n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1 \omega}{s^2 + \omega^2} + \frac{C_2 s}{s^2 + \omega^2}$$

Assume $0 < \zeta < 1$

$$+ \frac{C_3 \omega_d + C_4 (s + \sigma)}{(s + \sigma)^2 + \omega_d^2}$$

* 3 different ω 's *



$$X(s) = \frac{C_1 \omega}{s^2 + \omega^2} + \frac{C_2 s}{s^2 + \omega^2} + \frac{C_3 \omega_d + C_4 (s + \alpha)}{(s + \alpha)^2 + \omega_d^2}$$

Annotations:
 - Green arrow labeled 'row 6' points from $\frac{C_1 \omega}{s^2 + \omega^2}$ to $C_1 \sin(\omega t)$.
 - Green arrow labeled 'row 7' points from $\frac{C_2 s}{s^2 + \omega^2}$ to $C_2 \cos(\omega t)$.
 - Green arrow labeled 'row 8' points from $C_3 \omega_d$ to $C_3 \sin(\omega_d t)$.
 - Green arrow labeled 'row 9' points from $C_4 (s + \alpha)$ to $C_4 \cos(\omega_d t)$.

$$x(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t) + e^{-\alpha t} (C_3 \sin(\omega_d t) + C_4 \cos(\omega_d t))$$

Annotations:
 - A red arrow points from the $e^{-\alpha t}$ term down to the text $\alpha > 0 \quad (s > 0)$.
 - The text $\alpha > 0 \quad (s > 0)$ is written in blue.

as $t \rightarrow \infty$

$$x_{ss}(t) \approx C_1 \sin(\omega t) + C_2 \cos(\omega t) \rightarrow M \sin(\omega t + \phi)$$

↑
steady state

$$\sin(a+b) = \sin a \cos b + \cos a \sin b$$

$$\begin{aligned} \mu \sin(\omega t + \phi) &= \mu \sin(\omega t) \cos \phi + \mu \cos(\omega t) \sin \phi \\ &= C_1 \sin(\omega t) + C_2 \cos(\omega t) \end{aligned}$$

$$C_1 = \mu \cos \phi \quad C_2 = \mu \sin \phi$$

$$C_1^2 + C_2^2 = \mu^2 (\cos^2 \phi + \sin^2 \phi) \quad \tan \phi = \frac{C_2}{C_1}$$

$$\mu = \sqrt{C_1^2 + C_2^2}$$

$$\frac{C_2}{C_1} = \frac{\mu \sin \phi}{\mu \cos \phi}$$

For a sinusoidal input to a stable TF $G(s)$

$$X(s) = G(s) \cdot F(s) = G(s) \cdot \frac{\omega}{s^2 + \omega^2}$$

transient terms
die out

$$\left[G(s) \cdot \frac{\omega}{s^2 + \omega^2} = \frac{C_1 \omega}{s^2 + \omega^2} + \frac{C_2 s}{s^2 + \omega^2} + \text{other stuff} \right] \cdot (s^2 + \omega^2)$$

$$\left[G(s) \cdot \frac{\omega}{s^2 + \omega^2} = \frac{C_1 \omega}{s^2 + \omega^2} + \frac{C_2 s}{s^2 + \omega^2} + \text{other stuff} \right] \cdot (s^2 + \omega^2)$$

$$G(s) \omega = C_1 \omega + C_2 s + (s^2 + \omega^2) [\text{other stuff}]$$

$$G(j\omega) \cdot \cancel{\omega} = C_1 \cancel{\omega} + C_2 j\cancel{\omega} + \cancel{(-\omega^2 + \omega^2)} [\text{other stuff}] \Big|_{s=j\omega}$$

$$G(j\omega) = C_1 + C_2 j \rightarrow C_1 = \text{Real}(G(j\omega))$$

$$C_2 = \text{Imag}(G(j\omega))$$

→ we treat the input as unknown when finding the output

The transfer function is independent of the input.

$G(s)$ does not change based on $F(s)$



$$X(s) = G(s) \cdot F(s)$$